

Hsin-Wei (Aynslie) Lin

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Education

University of Washington

Sep 2026 - Jun 2028 (expected)

- M.S in Electrical and Computer Engineering

National Taiwan Ocean University

Sep 2022 – Jun 2026

- B.S in Electrical Engineering

Skills

- **Programming:** C, Python, MATLAB, G-code
- **Hardware & Embedded Systems:** Arduino Mega, NVIDIA Jetson Orin Nano, AMB82-mini, Motor Drivers, Stepper Motors, UART
- **Computer Vision:** YOLOv8, OpenCV, Dataset Annotation
- **Engineering Tools:** SolidWorks, Linux/Ubuntu

Research

Autonomous Dining Service Robot - Python / YOLOv8 / OpenCV / ArUco

Jul 2025 – Feb 2026

- Awarded the Undergraduate Research Grant by the National Science and Technology Council (NSTC).
- Designed a SolidWorks lift-based gripper mechanism for plate pickup.
- Collected and annotated a custom plate dataset for YOLOv8 after existing datasets performed poorly on plate viewed from the fixed camera angle; integrated the model with OpenCV for plate-state detection.
- Used plate position in the camera view as feedback for Proportional control (P-Control), adjusting Mecanum wheel motion during the final approach for more reliable plate positioning.
- Used ArUco marker to identify the dining location and guide the robot to the delivery position.

Projects

Intelligent CNC Plotter Machine - Python / G-Code / GRBL / CNC

Jul 2025 – Aug 2025

- Designed split-type SolidWorks screw-support structures to maintain lead-screw alignment and reduce mechanical binding during the motion.
- Wired A4988 motor-driver circuits with stepper motors and integrated OpenCV-based background removal into the team-developed Python GUI to simplify image processing before plotting.

Omnidirectional Mobile Robot Platform - Arduino Mega / UART / Sensors

Sep 2024 – Jan 2025

- Built a Mecanum-wheel mobile robot using Arduino Mega, L298N motor drivers, Bluetooth, and ultrasonic sensors.
- Implemented Bluetooth-based control and Mecanum-wheel motion using Arduino PWM, supporting lateral and diagonal movement.

Artificial Intelligence Doorbell - AMB82-mini / Arduino IDE / IFTTT

Feb 2024 – Jun 2024

- Integrated existing face recognition functionality with RTSP video stream on an AMB82-mini MCU.
- Integrated image transfer from an AMB82-mini edge device with Google Apps Script and Google Drive for cloud-based image storage.
- Connected Google Drive with LINE through IFTTT to automate image notifications following face-detection events.